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The research on cascading failure prediction in AC-DC hybrid power grids based on deep learning

Wenchao Qin[1*], Xufei Liu[1] and Yiran Tao[1]

*Correspondence:
Wenchao Qin
qinwc2025yedcc@outlook.com
[1]Power Dispatching and Control Center of Yunnan Power Grid Co., Ltd, Kunming 650011, Yunnan , China

Abstract

Accurate prediction of cascading faults in power grids can ensure system stability and prevent large-scale outages. Existing methods typically require learning effective statistical information from datasets with sufficient labels; however, in real-world scenarios, fault samples are scarce, leading to inaccurate fault identification. To address this, this paper presents a Hybrid Graph-Temporal Transformer (HGTT) to predict cascading faults in AC/DC hybrid power grids. The HGTT model integrates Graph Attention Networks (GAT) and Temporal Transformers, effectively capturing both spatial and temporal dependencies through an attention mechanism that accounts for electrical distances between nodes, as well as a causal attention-based temporal feature extraction module. Additionally, two self-supervised tasks are introduced to reduce reliance on labeled data. Experimental results show that HGTT achieves up to 14.8% higher accuracy than SVM, and reduces labeled data requirements by 50% under the self-supervised learning setting compared to fully supervised training.

Keywords Cascading faults, AC/DC hybrid power grid, Deep learning, Graph attention network, Temporal transformer

1 Introduction

The growing emphasis on renewable energy has driven a significant expansion in power generation from sustainable sources. Renewable energy is inherently intermittent and variable, necessitating advanced grid management strategies to ensure stable and reliable power supply [1]. In response to these challenges, AC/DC hybrid power grids have emerged as a pivotal solution. By integrating Alternating Current (AC) and Direct Current (DC) transmission systems, hybrid power grids enhance the capacity to efficiently distribute renewable energy, thereby improving overall grid resilience and flexibility [2]. One of the most critical issues in AC/DC hybrid power grids is the occurrence of cascading faults. Cascading failures are characterized by their rapid propagation, extensive impact, and high destructiveness, often leading to widespread power outages. A notable example is the 2016 South Australia blackout, where a local fault escalated into a

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large-scale cascading event, resulting in a significant disruption of power supply across the region [3]. Such incidents underscore the urgent need for accurate prediction and timely intervention to mitigate systemic risks and ensure the safe of power grids.

Physical model-based methods analyze fault propagation by establishing detailed dynamic models of the power system. For instance, Yan et al. proposed a propagation model based on a cascading fault tree [4], and Peng et al. constructed a fault evolution model considering electromechanical coupling characteristics [5]. While these methods provide a thorough understanding of fault mechanisms and can accurately describe fault propagation, they suffer from high modeling complexity. Determining accurate parameters is challenging, and the computational demands make these methods time-consuming, rendering them unsuitable for real-time prediction needs. Additionally, their sensitivity to changes in system parameters limits their adaptability to evolving grid conditions.

Statistical learning-based methods utilize historical data to mine fault propagation patterns. Eladl et al. employed Support Vector Machine (SVM) to establish a fault warning model [6], and Le et al. developed a fault risk assessment method based on random forests [7]. These methods offer fast computation speeds and are relatively easy to implement. However, they struggle to effectively extract high-dimensional nonlinear features inherent in power grid data. Moreover, they often neglect the underlying grid topology information, which is crucial for understanding fault propagation. Consequently, their prediction accuracy and generalization ability are limited, especially in dynamic grid environments.

Deep learning has gained traction in the power systems domain due to its ability to automatically extract features and model dependencies. Gong et al. introduced a grid fault diagnosis method based on Convolutional Neural Network (CNN) [8], Hossain et al. designed a fault prediction model leveraging Long Short-Term Memory (LSTM) networks [9], and Wang et al. utilized Graph Convolutional Networks (GCN) for fault propagation analysis [10]. Despite their promising performance, existing deep learning methods face significant challenges in simultaneously modeling static topological features and dynamic temporal features of the power grid. Additionally, these models require large amounts of labeled data for training, which is often scarce in real-world scenarios where fault events are rare.

Despite the advances reviewed above, three key research gaps remain unaddressed. First, existing graph-based models seldom embed physical propagation constraints into the attention mechanism. Second, simultaneous modeling of heterogeneous AC/DC topology and multi-scale temporal dynamics is rarely achieved within a unified framework. Third, the reliance on fully labeled fault data remains a persistent bottleneck due to the scarcity of cascading failure events.

To address these challenges, this paper proposes a novel hybrid deep learning model, called the Hybrid Graph-Temporal Transformer (HGTT), for predicting cascading failures in AC/DC hybrid power grids. The HGTT integrates Graph Attention Networks (GAT) and Temporal Transformers to effectively capture both spatial and temporal dependencies in the grid. GAT is particularly effective at modeling the topological relationships between nodes, as it allows the model to focus on relevant connections while dynamically adjusting attention weights. Meanwhile, the Temporal Transformer captures temporal dynamics by leveraging causal attention, enabling the identification of

long-term dependencies and early fault indicators over time. By combining these components, the HGTT addresses the limitations of existing approaches, which often struggle to capture both static topology and evolving fault dynamics. The HGTT model is validated on IEEE standard test cases as well as real-world data from China's power grid. The results show that, compared to SVM, HGTT achieves a relative accuracy improvement of over 15% while requiring only 50% of the labeled data used in fully supervised methods.

2 Related work

The prediction of cascading failures in AC/DC hybrid power grids can be broadly categorized into three primary approaches.

Physical model-based methods aim to analyze the fault propagation mechanism by establishing dynamic models of the power system. Yan et al. [4] proposed a large-scale early-warning system for AC/DC hybrid power grids that integrates decision trees with physical models. Their approach combines semi-supervised deep learning techniques with a decision tree framework to enhance the accuracy of early fault detection and improve the system's adaptability. Peng et al. [5] focused on real-time stability in LCC-HVDC (Line Commutated Converter High Voltage Direct Current) systems and proposed an AC/DC hybrid power system damping control method based on estimated model predictive control. Their work emphasizes the importance of real-time system adjustments to enhance dynamic performance and mitigate cascading fault risks. Dong et al. [11] conducted extensive simulations of cascading failures in AC/DC hybrid grids and investigated fault propagation dynamics, contributing valuable insights into the failure mechanisms of complex grid systems. Lambrichts et al. [12] introduced a linear recursive state estimation method, utilizing synchronous phasor measurements to improve the accuracy of state estimation in real-time operations. Karaca et al. [16] analyzed the Turkish power grid as a case study to explore the characteristics and operational challenges of transmission systems, providing essential references for the planning and operation of large-scale hybrid power grids. These methods typically involve high modeling complexity, difficulty in parameter determination, and long computation times. Furthermore, they struggle to meet real-time prediction requirements, are sensitive to system parameter changes, and exhibit poor adaptability to dynamic grid environments.

Statistical learning-based methods leverage historical data to identify patterns and mine fault propagation laws. These methods do not require detailed physical modeling and can often adaptively learn the fault propagation mechanisms from data, offering advantages in computation speed and practical deployment. Eladl et al. [6] proposed an Internet of Things (IoT)-based protection scheme using SVM to enhance fault protection and restoration in multi-terminal HVDC transmission networks. Their method demonstrated a significant improvement in fault detection and mitigation for DC transmission grids. Le et al. [7] developed a fault identification method for DC distribution systems using random forest-based prediction models, which improved the accuracy of detecting series arc faults. Zaben et al. [14] reviewed machine learning methods, providing valuable insights into selecting the most appropriate techniques for specific grid configurations. Bramareswara et al. [15] used a combination of discrete wavelet transforms and deep neural networks to detect and classify faults in multi-regional grid-connected

microgrids, with applications to hybrid energy systems. While statistical learning-based methods offer fast computation and are easier to deploy in practice, they also face challenges. One of the main limitations is their inability to effectively capture high-dimensional nonlinear features in power grid data, which restricts their predictive accuracy and generalization capability. Another limitation is the dependence on high-quality historical data, which can be scarce, particularly for rare fault events.

Deep learning-based methods, particularly CNN, recurrent neural networks (RNN), and GCN, have shown promise in fault diagnosis and prediction tasks. Gong et al. [8] proposed an improved CNN-based fault diagnosis method for DC-DC inverters, utilizing global average pooling and two-dimensional feature maps to enhance diagnostic accuracy. Seyedi et al. [13] developed a data-driven hybrid AC-DC microgrid voltage stability prediction method, leveraging deep learning to improve post-fault voltage stability predictions. Wang et al. [10] used GCN to identify power transmission system parameters, demonstrating the potential of graph-based models to capture network topology features and improve fault propagation analysis. Hu et al. [17] investigated the operation and coordinated control of integrated hybrid distribution transformers under grid fault conditions, proposing methods to enhance the system's fault adaptability and improve fault tolerance. Deep learning methods have several advantages: they can automatically extract features, reduce the need for manual feature engineering, and handle complex nonlinearities in the data. However, deep learning approaches also face notable challenges. One major issue is their difficulty in modeling both static topological features and dynamic temporal features simultaneously.

In summary, the above methods leave three key gaps: physical topology constraints are seldom incorporated into graph attention mechanisms; joint modeling of heterogeneous AC/DC structures and temporal dynamics within a unified framework is still lacking; and the scarcity of fault-event labels is inadequately addressed by purely supervised approaches. Recent work has begun to tackle related issues in structural health monitoring and reliability contexts [22, 23, 24]; however, these limitations persist in power-grid cascading failure prediction.

3 Design of HGTT

In this study, we introduce HGTT, as illustrated in Fig. 1. The architecture seamlessly integrates a GAT with a Transformer module, further enhanced by self-supervised learning tasks to improve feature representation. This integration allows the model to effectively capture both spatial and temporal dependencies within the power grid, thereby providing accurate and reliable fault predictions.

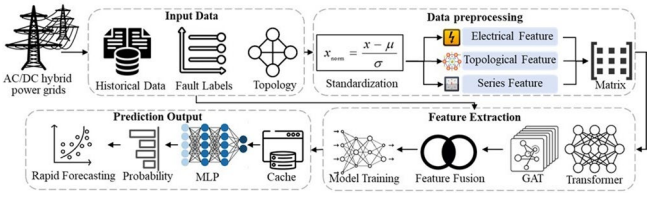


Fig. 1 Hybrid graph-temporal transformer network architecture

3.1 Hybrid deep learning model

The architecture integrates a GAT [18] with a Temporal Transformer module, augmented by self-supervised learning tasks to enhance feature representation (e.g. Figure 2). We model the AC/DC hybrid power grid as a heterogeneous graph $G = (V, E)$, where $V = \{v_1, v_2, \ldots, v_n\}$ denotes the set of nodes, comprising AC bus nodes V_{AC}, DC converter nodes V_{DC}, Transformer nodes V_T. $E = \{e_{ij}\}$ represents the set of edges, describing the physical connections between nodes. The node feature matrix $X \in \mathbb{R}^{N \times F}$ encapsulates both electrical and temporal features, where N is the total number of nodes and F is the feature dimension. Specifically, for each node Electrical Quantities such as Voltage amplitude U_i, phase angle θ_i, active power P_i, reactive power Q_i. Topological features include node type, connectivity, and electrical distance. Temporal features consist of historical measurement data sequences. The core of our model leverages the GAT to capture the spatial dependencies within the power grid. The graph attention layer updates the hidden state of each node v_i at layer $l + 1$ based on its neighboring nodes $\mathcal{N}_i$ as follows:

$$h_i^{(l+1)} = \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} W^{(l)} h_j^{(l)}\right) \tag{1}$$

where $h_i^{(l)} \in \mathbb{R}^F$ is the hidden state of node v_i at layer l+1, $W^{(l)} \in \mathbb{R}^{F' \times F}$ is the learnable weight matrix, $\sigma(\cdot)$ is the LeakyReLU activation function, $\mathcal{N}_i$ denotes the neighborhood set of node v_i, α_{ij} is the attention coefficient between nodes v_i and v_j. The attention coefficient α_{ij} is computed to incorporate an electrical distance perception mechanism, enhancing the model's ability to account for the physical properties of the power grid:

$$\alpha_{ij} = \frac{\exp\left(\mathrm{LeakyReLU}\left(\mathbf{a}^\top \left[W^{(l)} h_i^{(l)} \parallel W^{(l)} h_j^{(l)}\right]\right)\right)}{\sum_{k \in \mathcal{N}_i} \exp\left(\mathrm{LeakyReLU}\left(\mathbf{a}^\top \left[W^{(l)} h_i^{(l)} \parallel W^{(l)} h_k^{(l)}\right]\right)\right)} \cdot f(Z_{ij}) \tag{2}$$

where $\mathbf{a} \in \mathbb{R}^{2F'}$ is the attention vector parameter, $\parallel$ denotes the concatenation operation, $f(Z_{ij}) = \beta \cdot \exp(-\gamma Z_{ij})$ is the electrical distance weight function, Z_{ij} represents the electrical distance between nodes v_i and v_j, β and γ are adjustable hyperparameters. The electrical distance Z_{ij} is calculated by considering impedance values, topological connections, and power transmission capacities between nodes. The electrical distance [21] is a metric that quantifies the physical relationship between two nodes in the grid

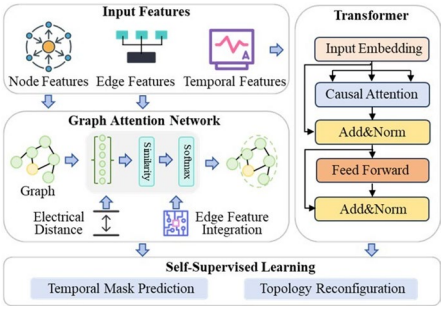


Fig. 2 The hybrid deep learning architecture

based on impedance, topological connections, and power transmission capacities. Concretely, the electrical distance between nodes v_i and v_j is computed as:

$$d_{ij} = \frac{|Z_{ij}|}{C_{ij}^{\beta} \cdot P_{ij}^{\gamma}} \tag{3}$$

where $|Z_{ij}|$ is the magnitude of the equivalent impedance, C_{ij} is the normalised transmission capacity of the path, and P_{ij} is the normalised active power flow. The hyperparameters β and γ control the relative weights of capacity and power-flow terms, respectively; they were set to $\beta = 0.7$ and $\gamma = 0.3$ via grid search. The electrical distance is computed offline from the steady-state admittance matrix and is not updated online during operation. This mechanism embeds physical constraints into the attention mechanism, ensuring that the model prioritizes more physically significant connections. To enhance the model's expressiveness and capture diverse interaction patterns, we employ *k* independent attention heads. The updated hidden state with multi-head attention [19] is given by:

$$h_i^{(l+1)} = \sigma\left(\frac{1}{K}\sum_{k=1}^{K}\sum_{j\in\mathcal{N}_i}\alpha_{ij}^{k}W^{(l,k)}h_j^{(l)}\right) \tag{4}$$

Where α_{ij}^{k} is the attention coefficient for the *k*-th attention head, $W^{(l,k)} \in \mathbb{R}^{F'\times F}$ is the learnable weight matrix in layer *l*. The multi-head mechanism capturing a richer set of interactions between nodes. To fully utilize the edge feature information inherent in power systems, we incorporate edge features into the attention mechanism:

$$e_{ij} = W_e\left[h_i \parallel h_j \parallel x_{ij}\right] \tag{5}$$

where x_{ij} is the feature vector of edge e_{ij}, $W_e \in \mathbb{R}^{d_e\times(2F'+F_e)}$ is the weight matrix for edge feature fusion, d_e is the dimension of the fused edge features. This integration ensures that the GAT effectively captures the complex interactions between nodes by considering both node and edge features, thereby providing robust feature representations for subsequent fault prediction tasks. To capture the temporal dynamics of the power grid system, we incorporate a Temporal Transformer module based on the causal attention mechanism. This module processes the time series data sequences $X = \{x_1, x_2, \ldots, x_T\}$ to model temporal dependencies. The input time series data *X* is encoded into query (*Q*), key (*k*), and value (*V*) matrices using learnable transformation matrices [20]:

$$Q = XW_Q, K = XW_K, V = XW_V \tag{6}$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d_{\text{model}}\times d_k}$ are learnable transformation matrices, d_{model} is the dimension of the model, d_k is the dimension of the attention space. To ensure that the model only attends to past and present information, a causal mask *M* is applied during the self-attention calculation:

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}^{\circ} M\right)V \tag{7}$$

where $^{\circ}$ denotes element-wise multiplication, and *M* is defined as:

$$M_{ij} = \begin{cases} 0, & \text{if } i < j, \\ 1, & \text{if } i \geqslant j. \end{cases} \tag{8}$$

This masking ensures that the prediction for a given time step i only depends on time steps $j \leqslant i$, thereby preserving the temporal causality required for accurate time series modeling. Each encoder layer within the Temporal Transformer comprises a multi-head causal self-attention sub-layer followed by a feed-forward neural network sub-layer. Layer normalization and residual connections are applied to facilitate stable and efficient training:

$$h^l = \text{LayerNorm}\left(x^l + \text{MultiHead}(Q, K, V)\right) \tag{9}$$

$$x^{l+1} = \text{LayerNorm}\left(h^l + \text{FFN}(h^l)\right) \tag{10}$$

where FFN denotes the feed-forward network, LayerNorm is the layer normalization function, h^l and x^{l+1} represent the hidden states at layer l and $l+1$, respectively. This architecture effectively captures temporal dependencies in the grid's operational data, enhancing the model's ability to predict cascading failures by learning complex temporal patterns. To further improve the feature representation capabilities of the model, we incorporate two self-supervised learning tasks: Temporal Mask Prediction and Topological Reconstruction. Specifically, for temporal mask prediction, a random masking strategy is applied with a mask ratio of 15% of all time steps in each window. Masked positions are selected independently and uniformly at random;. For topological reconstruction, the model reconstructs the full graph adjacency matrix), using the binary cross-entropy loss defined below. The self-supervised pre-training phase uses 100% of the available training samples without fault labels; only the subsequent fine-tuning phase requires fault labels. The loss function for temporal mask prediction is defined as:

$$\mathcal{L}_{\text{mask}} = \frac{1}{N_m} \sum_{i=1}^{N_m} \|x_i - \hat{x}\|_{i2}^2 \tag{11}$$

where N_m is the number of masked time steps, x_i is the original value at time step i, $\hat{x}_i$ is the model's prediction for the masked value. This task encourages the model to learn robust temporal representations by reconstructing missing information, thereby enhancing its ability to capture underlying temporal patterns in the data. This task aims to reconstruct the adjacency matrix of the graph structure, thereby encouraging the model to learn the underlying topological relationships within the power grid. The loss function for topological reconstruction is given by:

$$\mathcal{L}_{\text{topo}} = -\frac{1}{|E|} \sum_{(i,j)\in E} \left[A_{ij} \log(\hat{A}_{ij}) + (1 - A_{ij}) \log(1 - \hat{A}_{ij}) \right] \tag{12}$$

where A_{ij} is the actual adjacency matrix element between nodes v_i and v_j, $\hat{A}_{ij}$ is the predicted adjacency matrix element, $|E|$ is the total number of edges. This task ensures that the model accurately captures the physical connections between nodes, enhancing the overall feature representation by embedding topological information into the model. The overall learning objective integrates the self-supervised tasks with the fault

prediction task, forming a multi-task learning framework. The combined loss function is formulated as:

$$\mathcal{L} = \mathcal{L}_{\text{mask}} + \lambda \mathcal{L}_{\text{topo}} + \gamma \mathcal{L}_{\text{pred}} \tag{13}$$

where $\mathcal{L}_{\text{pred}}$ is the cross-entropy loss for the fault prediction task, λ and γ are weight coefficients that balance the contributions of the respective loss components. This multi-task learning approach ensures that the model not only predicts faults accurately but also learns meaningful representations of both temporal dynamics and topological structures within the AC/DC hybrid power grid. The key parameters in the HGTT model were carefully selected through extensive ablation studies. For the GAT component, the number of GAT layers was set to 3 to balance model depth and computational efficiency, while the number of attention heads (K) was set to 8, based on empirical studies showing optimal performance. The hidden dimensions were configured as [64, 128, 256] to gradually increase the feature representation capacity. For the Temporal Transformer component, the number of encoder layers was set to 4, and the time window size was chosen to be $T = 24$ time steps. Given the 15-minute sampling interval of the IEEE-39 dataset, this corresponds to a look-back window of 6 h, which was determined by autocorrelation analysis of node-state sequences: the autocorrelation coefficient drops below 0.1 beyond lag 24, indicating that earlier observations contribute negligible information. The attention dimension was set to 512 to ensure sufficient capacity for temporal feature extraction. Additionally, the weight coefficients α and β were set to 0.3 and 0.7, respectively, through a grid search process. During the training of the HGTT model, several key challenges arose. The combination of GAT and Transformer led to unstable training dynamics and gradient issues. To address this, gradient clipping was applied with a threshold of 5.0, and a warm-up learning rate schedule was used, gradually increasing the learning rate from 1e-5 to 1e-3 over the first 1000 iterations. The model also showed signs of overfitting due to limited fault samples. To mitigate this, self-supervised pre-training tasks were implemented, and dropout (rate = 0.1) was applied in both GAT and Transformer modules. Additionally, early stopping with a 10 epochs was employed. The integration of these tasks facilitates the extraction of comprehensive features, thereby enhancing the model's overall predictive performance.

3.2 Cascading failure prediction algorithm

Building upon the hybrid architecture of GAT and Temporal Transformers, this section delineates the specific implementation process of the cascading failure prediction algorithm. The algorithm leverages historical operational data and the power grid's topology to predict the probability of faults occurring within the AC/DC hybrid power grid. The cascading failure prediction algorithm is designed to process input datasets comprising historical operational data and the power grid's topology to output fault probability predictions for each node in the grid. Formally, the algorithm operates as follows. Input historical operation dataset $D = \{(X_t, y_t)\}_{t=1}^{T}$, where X_t represents the operational state at time *t*, and y_t denotes the corresponding fault labels. Power grid topology structure $G = (V, E)$, where $V = \{v_1, v_2, \ldots, v_n\}$ is the set of nodes and $E = \{e_{ij}\}$ is the set of edges. Output is fault probability prediction vector $p \in [0, 1]^N$, where N is the total number of nodes. To ensure the effectiveness of the prediction model, the input data undergoes standardization and feature construction. Standardizing the input features is

crucial for stabilizing and accelerating the training process. The standardized input data x_{norm} is computed as:

$$x_{\text{norm}} = \frac{x - \mu}{\sigma} \tag{14}$$

where μ and σ are the mean and standard deviation of the feature values, respectively. The node feature matrix $X \in \mathbb{R}^{N \times F}$ is constructed by concatenating electrical, topological, and temporal features:

$$X = [X_{\text{elec}} \parallel X_{\text{topo}} \parallel X_{\text{temp}}] \tag{15}$$

where X_{elec} represents electrical quantity characteristics such as voltage, current, and power. X_{topo} encompasses topological characteristics including node type and connectivity. X_{temp} includes temporal characteristics derived from historical measurement data. To capture the spatial dependencies within the power grid, the algorithm employs an *L*-layer GAT to extract static node representations. The static representation after *L* layers is denoted as $H^{(L)} \in \mathbb{R}^{N \times F}$ and is computed as:

$$H^{(L)} = \text{GAT}^{(L)}(H^{(L-1)}, A, X), \quad l = 1, 2, \ldots, L \tag{16}$$

where $H^{(0)} = X$ is the initial node feature matrix. *A* is the adjacency matrix of the graph *G*. $\text{GAT}^{(l)}$ denotes the GAT layer at the *l*-th level. The final static node representation $H^{(L)}$ encapsulates the spatial interactions and structural information of the power grid. To model the temporal dynamics of the power grid, the algorithm integrates a Temporal Transformer module that processes time series data within a specified time window *w*. For each node v_i, the temporal dynamic features S_t^i at time *t* are encoded as:

$$S_t^i = \text{Transformer}(X_{t-w:t}) \tag{17}$$

where $X_{t-w:t}$ denotes the sequence of input features from time $t - w$ to *t*. The Transformer module captures the temporal dependencies and patterns essential for accurate fault prediction. To synergize the static and dynamic features, a cross-modal attention fusion layer is designed. For each node v_i, the fused feature Z_i is computed as:

$$Z_i = W_f \left[H_i^{(L)} \parallel g(H_i^{(L)}, S_t^i) \right] \tag{18}$$

where W_f is the learnable weight matrix for fusion. $g(\cdot)$ is the attention fusion function defined as:

$$g(H_i^{(L)}, S_t^i) = \sum_{j=1}^{w} \alpha_j S_{t-j}^i \tag{19}$$

where α_j are the attention weights computed as:

$$\alpha_j = \frac{\exp(W_a[H_i^{(L)} \parallel S_{t-j}^i])}{\sum_{k=1}^{w} \exp(W_a[H_i^{(L)} \parallel S_{t-k}^i])} \tag{20}$$

where W_a is the weight matrix for attention weight calculation. This fusion mechanism allows the model to effectively combine static structural information with temporal

dynamic data, enhancing the feature richness for fault prediction. The fused feature Z_i is then passed through a Multi-Layer Perceptron (MLP) to generate the fault probability p_i for each node v_i. The MLP consists of multiple fully connected layers defined as:

$$\mathrm{MLP}(Z_i) = W_2 \cdot \mathrm{ReLU}(W_1 Z_i + b_1) + b_2 \tag{21}$$

where W_1 and W_2 are learnable weight matrices, b_1 and b_2 are bias vectors, ReLU is the Rectified Linear Unit activation function. This architecture enables the model to produce accurate fault probability predictions by leveraging both spatial and temporal information. To enhance computational efficiency and focus on the most significant interactions, critical attention connections are selected through importance sampling. The refined attention coefficient $\tilde{\alpha}_{ij}$ is defined as:

$$\tilde{\alpha}_{ij} = \begin{cases} \alpha_{ij}, & \text{if } \alpha_{ij} > \tau, \\ 0, & \text{otherwise}, \end{cases} \tag{22}$$

where τ is an adaptive threshold calculated as:

$$\tau = \mu + \beta\sigma \tag{23}$$

with μ and σ are the mean and standard deviation of the attention weights α_{ij}, β is an adjustable parameter. This selective attention mechanism ensures that only the most relevant connections contribute to the prediction, thereby reducing computational overhead without compromising prediction accuracy. To facilitate rapid and efficient predictions, a node feature cache update mechanism is implemented. The cached feature C_t^i for node v_i at time *t* is updated as:

$$C_t^i = \gamma C_{t-1}^i + (1-\gamma) h_t^i \tag{24}$$

where C_t^i is the cached feature of node $.v_i.$ at time *t*, $h_t^i s$ is the node feature at the current time step, $\gamma \in [0,1]$ is the cache update rate. The cached feature is subsequently used for rapid fault probability prediction:

$$\hat{p}_i = \sigma(W_c C_t^i + b_c) \tag{25}$$

where W_c is the weight matrix for cache feature transformation, b_c is the bias vector. This caching mechanism enhances prediction speed by maintaining an up-to-date representation of node features, enabling real-time fault probability estimations. However, the cache relies on a fixed update rat calibrated under quasi-static topology conditions. During rapid topology changes, the cached features may lag behind actual system states, potentially reducing prediction accuracy.

4 Experiment and results

4.1 Experiment setup

Dataset The experimental validation uses the IEEE-39 node standard grid and a real-world case from the China Power Grid. The IEEE-39 standard test case is an AC network, constructed by replacing two existing AC transmission corridors with point-to-point LCC-HVDC links. Each DC link is modeled as a controlled current source at the rectifier bus and a controlled voltage source at the inverter bus. DC-side node features include DC voltage, DC current, and converter firing angle; these three features

replace the reactive-power component in the AC node feature vectors, yielding a unified 12-dimensional feature vector for all node types via zero-padding for absent features. AC-side and DC-side measurements are aligned at the same 15-minute sampling epoch, featuring 39 AC buses, 2 DC lines, and 10 synchronous generators with a total installed capacity of 6.2 GW. It also includes 46 AC lines and 19 load centers, with a total load of approximately 6.1 GW. The data, collected from January 2020 to December 2023 at a 15-minute sampling frequency, include 12 features per node such as voltage, current, power, and stability indices. The dataset contains 534 cascading fault events (including equipment-level, N-1, and N-k faults), split into 427 training and 107 test events. The China Power Grid dataset is a real-world dataset with 126 bus nodes, 189 AC lines, and 15 DC lines operating at higher voltages (±800 kV/±500 kV). The system includes 78 transformers and 24 sets of shunt capacitors and reactors. Data collected from March 2021 to December 2023 at 5-minute intervals include 16 features per node, such as electrical quantities (voltage, current, power) and dynamic features (stability indices and impedance trajectory parameters). The dataset includes 312 cascading failure events categorized by equipment-level, system-level, and regional-level failures, split into 252 training and 60 test events. Regarding class imbalance, in the IEEE-39 dataset, fault-state samples represent approximately 18.3% of all time steps, yielding a positive-to-negative ratio of roughly 1:4.5. In the China Power Grid dataset, the ratio is approximately 1:6.2. The evaluation metrics include Accuracy, Precision, Recall, F1-score, and AUC; AUC is additionally reported as a threshold-independent metric.

Experimental Environment The hardware configuration includes an Intel Xeon Gold 6248R CPU, an NVIDIA RTX 4080 GPU, 256 GB of DDR4 memory, and a 2 TB NVMe SSD for fast data access and storage. The software environment consists of Ubuntu 20.04 LTS as the operating system, PyTorch 1.12.0 as the deep learning framework, CUDA version 11.6 for GPU acceleration, and Python 3.8.10 for programming. Model Configuration: The GAT module consists of 3 layers, with hidden layer dimensions of [64, 128, 256], and 8 attention heads. The activation function used is LeakyReLU with an alpha value of 0.2, and a dropout rate of 0.1 is applied to prevent overfitting. The Temporal Transformer module consists of 4 encoder layers, each with 8 self-attention heads and a feedforward network dimension of 512. The time window size is set to 24, and layer normalization with a ϵ value of 1e-5 is applied to stabilize training. The training process uses a batch size of 64 and a learning rate of 0.001, optimized using the Adam optimizer with $\beta_1 = 0.9$ and$\beta_2 = 0.999$. The loss function is weighted, with the topology reconstruction loss set to 0.3 and the prediction loss to 0.7.

Comparison Method For performance comparison, the proposed method is evaluated against several typical approaches, including SVM for fault warning, Random Forest (RF) for risk assessment, CNN for fault diagnosis, LSTM, Transformer for fault prediction, and GCN for fault analysis.

5 Result

Table 1 presents a comparative analysis of prediction performance across benchmark methods. Traditional statistical methods such as KNN, SVM, and RF exhibit relatively poor performance, with accuracies ranging from 73% to 82%. KNN achieves the lowest accuracy due to its inability to handle nonlinear relationships in power grid data. Similarly, SVM and RF also show subpar performance, primarily because they cannot

Table 1 The comparative analysis of the prediction performance

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC (%)
IEEE-39 node standard dataset					
SVM	78.42 ± 0.91	75.67 ± 1.03	76.25 ± 0.86	75.94 ± 0.92	81.23 ± 0.73
RF	80.47 ± 0.94	79.86 ± 0.92	80.62 ± 0.93	80.14 ± 0.79	84.52 ± 0.89
KNN	79.28 ± 0.95	80.82 ± 1.12	79.16 ± 0.91	79.98 ± 0.88	81.45 ± 0.82
CNN	85.36 ± 0.79	83.29 ± 1.07	84.17 ± 0.65	83.62 ± 0.92	87.82 ± 0.75
LSTM	87.68 ± 0.72	85.80 ± 0.81	86.37 ± 0.72	86.16 ± 0.85	89.47 ± 0.78
GCN	88.94 ± 0.65	87.26 ± 0.77	87.83 ± 0.69	87.50 ± 0.82	90.64 ± 0.68
Transformer	90.37 ± 0.78	89.36 ± 0.84	89.57 ± 0.73	89.46 ± 0.68	92.29 ± 0.74
HGTT	**92.79 ± 0.49**	**91.46 ± 0.78**	**91.81 ± 0.48**	**91.68 ± 0.64**	**94.39 ± 0.57**
China power grid dataset					
SVM	75.26 ± 0.89	72.84 ± 1.29	73.51 ± 0.95	73.16 ± 0.89	78.69 ± 0.83
RF	79.58 ± 0.81	77.13 ± 1.04	77.81 ± 0.83	77.42 ± 0.72	82.37 ± 0.91
KNN	80.23 ± 0.90	71.26 ± 1.21	75.94 ± 0.77	73.52 ± 0.74	80.82 ± 0.82
CNN	82.76 ± 0.92	80.44 ± 0.92	81.26 ± 0.71	80.83 ± 0.75	85.94 ± 0.87
LSTM	84.93 ± 0.90	82.86 ± 0.80	83.57 ± 0.82	83.16 ± 0.82	87.26 ± 0.67
GCN	86.35 ± 0.78	84.57 ± 0.82	85.13 ± 0.79	84.83 ± 0.66	88.72 ± 0.58
Transformer	88.41 ± 0.82	87.25 ± 0.76	87.14 ± 0.69	87.48 ± 0.69	90.16 ± 0.53
HGTT	**90.06 ± 0.64**	**89.24 ± 0.73**	**89.31 ± 0.68**	**88.59 ± 0.71**	**92.17 ± 0.49**

effectively capture deep, high-dimensional nonlinear features. Additionally, these methods demonstrate limited generalization when confronted with dynamic data in power grid systems. CNNs achieve modest improvements and excel at extracting local features through convolution operations; however, they struggle to handle the irregular topological structures of power grids. LSTM, designed for time-series modeling, is effective in capturing temporal dependencies, but its sequential nature limits parallelization and its ability to model spatial structural features, while long-term dependency issues may affect the recognition of early fault indicators. GCNs achieve higher accuracies by leveraging graph structures to model node relationships, but they overlook the impact of electrical distance on fault propagation. In contrast, HGTT improves performance over SVM by 14.37% on the IEEE-39 dataset and 14.80% on the China Power Grid dataset. Compared to the Transformer, the absolute gain is 2.42% on IEEE-39 and 1.65% on the China Power Grid dataset. The performance improvement is attributed to the electrical distance perception mechanism, which reflects fault propagation in physical space, and the causal attention mechanism, which captures temporal dependencies while preventing information leakage.

Table 2 delineates the prediction performance across different fault types. The HGTT model achieves the highest accuracy of 94.23% for single device faults. This high performance is attributed to the straightforward fault propagation path, pronounced system response characteristics, and ample historical samples, which collectively facilitate effective model learning. The electrical distance perception mechanism further enables precise localization of the fault source, enhancing prediction reliability. For N-1 faults, the HGTT model maintains strong performance with an accuracy of 91.96%. These faults involve interactions between multiple devices, introducing complexity in fault propagation. The graph attention mechanism effectively captures the coupling relationships between devices, while the causal attention mechanism accurately identifies the sequence of fault propagation, thereby sustaining high prediction accuracy. N-k faults represent the most challenging category, with the HGTT model achieving an accuracy of 89.57%. The complexity arises from multiple fault couplings leading to intricate system

Table 2 The prediction performance across different fault types

Fault types	Method	Accuracy	Recall	Forecast lead time (min)
Single device faults	LSTM	89.46 ± 0.72	88.61 ± 0.62	17.26 ± 1.48
	GCN	90.81 ± 0.69	90.12 ± 0.74	18.49 ± 1.51
	Transformer	92.17 ± 0.62	91.52 ± 0.65	20.94 ± 1.27
	HGTT	**94.23 ± 0.48**	**93.70 ± 0.54**	**23.17 ± 1.13**
N-1 faults	LSTM	86.72 ± 0.78	85.94 ± 0.83	14.87 ± 1.69
	GCN	88.36 ± 0.84	87.56 ± 0.90	16.24 ± 1.46
	Transformer	90.21 ± 0.75	89.56 ± 0.72	18.69 ± 1.39
	HGTT	**91.96 ± 0.66**	**91.28 ± 0.74**	**20.46 ± 1.27**
N-k faults	LSTM	84.26 ± 0.91	83.45 ± 0.81	12.56 ± 1.53
	GCN	85.93 ± 0.82	85.18 ± 0.92	13.84 ± 1.27
	Transformer	87.03 ± 0.79	87.13 ± 0.75	15.73 ± 1.38
	HGTT	**89.57 ± 0.76**	**88.92 ± 0.73**	**18.26 ± 0.95**

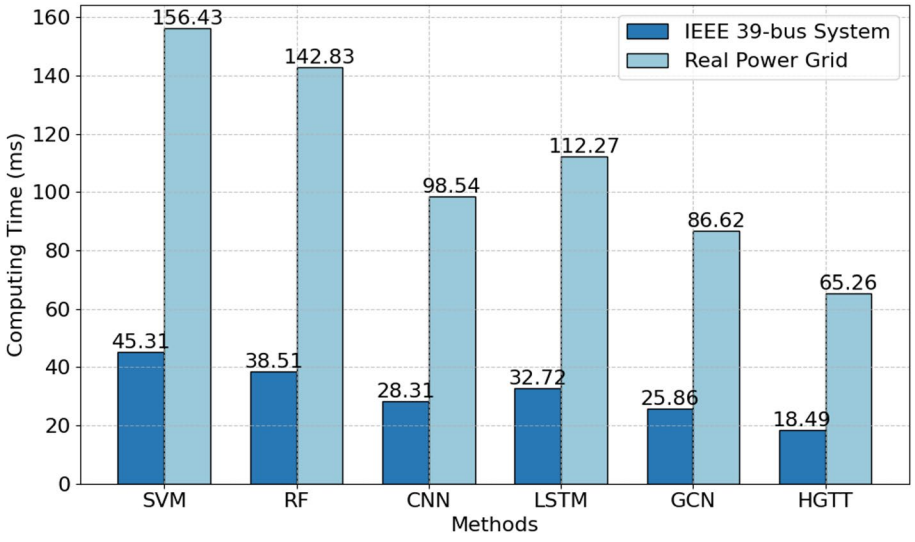


Fig. 3 Computational efficiency comparison

responses and a scarcity of samples, which complicates the learning process. Nonetheless, the HGTT model sustains commendable performance due to its robust modeling of multi-point fault propagation through graph structures and the utilization of self-supervised learning, which mitigates the issue of limited samples. The multi-head attention mechanism further provides redundant feature protection, enhancing the model's resilience to complex fault scenarios.

Figure 3 illustrates the computational efficiency of the HGTT model compared to traditional methods. The chart shows the computation time (in milliseconds) for each method across two datasets. The results highlight that the HGTT model achieves lower inference latency than simpler baselines such as SVM and CNN. Specifically, the average per-sample inference time is reduced by approximately 52.9% relative to the mean inference time of SVM, RF, CNN, LSTM, and GCN on the same hardware.This counterintuitive advantage over simpler models is due to the sparse attention mechanism, which reduces the effective complexity from $O(N^2)$ to $O(N \log N)$per layer by operating only on attention weights exceeding an adaptive threshold, as well as GPU-based batch parallelism, which benefits HGTT's tensor operations more than the CPU-bound SVM and tree-based RF. It should be noted that this comparison reflects online inference only

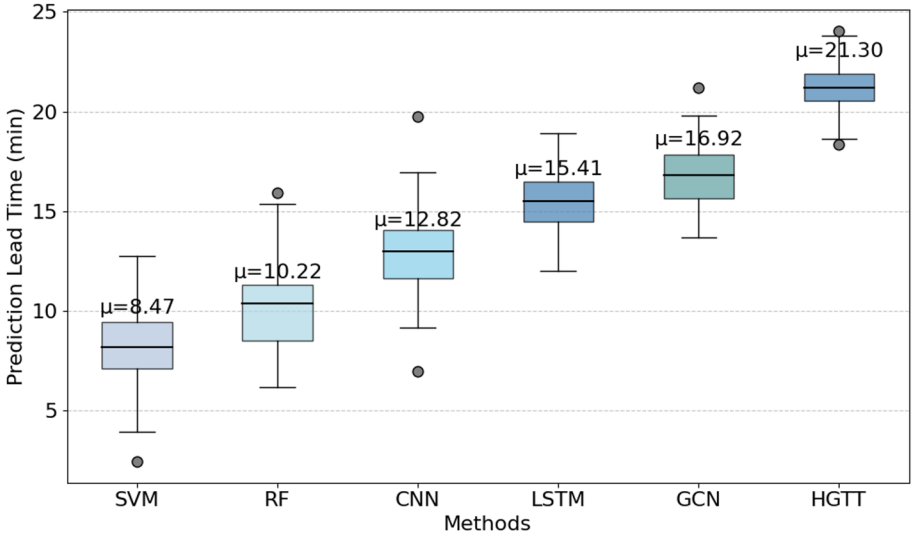


Fig. 4 Comparison of prediction lead time

Table 3 The results of ablation experiments

Method	Accuracy	Recall	F1-score	Calculation time(ms)
HGTT	92.79	91.81	91.68	18.57
Electrical distance(w/o)	89.42	88.45	88.38	17.82
Causal attention(w/o)	88.18	87.94	87.61	16.92
Self-supervised learning(w/o)	87.51	86.54	86.29	18.36
Attention mechanism(w/o)	91.93	91.51	90.94	28.75

and does not include training time, where HGTT is substantially more computationally expensive than the baselines.

Figure 4 presents a comparison of the prediction lead time (in minutes) for each method. The data shows that the HGTT model demonstrates a significant increase in lead time, with an average of 21.30 min for the real power grid. This is attributed to multi-scale feature fusion, which provides a comprehensive representation of the system state, and the causal attention mechanism, which improves the model's ability to recognize early fault indicators. The incorporation of graph structure features further enhances the model's ability to predict fault propagation trends in advance. This extended lead time is crucial for timely predictions, enabling the implementation of preventive measures and mitigating the impact of cascading failures on power grid stability.

Table 3 shows the accuracy, recall, F1-score, and computation time (in milliseconds) for the full HGTT model and various ablation configurations. The incorporation of the electrical distance perception mechanism leads to a performance improvement of 3.37%, as it introduces physical constraints that enhance prediction reliability and help identify potential fault propagation paths. The causal attention mechanism contributes a further 4.61% improvement, ensuring that predictions rely solely on historical data, increasing sensitivity to early fault indicators, and reducing false correlations. The integration of self-supervised learning tasks results in a 5.28% performance boost, improving the model's ability to handle limited labeled data and enhancing robustness to noise and missing values. Without these tasks, the model's accuracy drops to 87.51%. Finally, removing the attention mechanism leads to a performance decline, with accuracy reduced to 91.93%.

Notably, removing the attention mechanism increases computation time from 18.57 ms to 28.75 ms. This anomaly arises because the sparse attention mechanism in HGTT uses an adaptive importance-sampling threshold to prune irrelevant node pairs; in the ablated variant, attention is replaced by dense mean aggregation over all neighbors, eliminating sparsity and requiring full neighborhood traversal. Additionally, the absence of structured attention outputs forces the downstream MLP to process a higher-dimensional concatenated feature vector, increasing latency. Thus, the increase in computation time is not due to added complexity, but rather the loss of the sparse computation advantage of the original attention module.

Figure 5 examines the robustness of various prediction methods under increasing noise levels, ranging from 0% to 25%. The chart shows the prediction accuracy (%) for each method as noise levels increase. Traditional methods, such as SVM, RF, CNN, and LSTM, exhibit significant performance degradation even under low noise conditions (up to 5%), primarily due to their limited feature representation capabilities and inability to leverage contextual information for error correction. Among the deep learning methods, CNNs and LSTMs experience sharp declines in performance once noise levels exceed 15%, with accuracies dropping to around 79.3% for LSTM and 72.5% for CNN at high noise levels. GCNs maintain a moderate level of performance up to 20% noise, with accuracy levels around 84.9%, but still suffer a noticeable decrease in accuracy under higher noise conditions. In contrast, the proposed HGTT model demonstrates the highest robustness, maintaining an accuracy of over 83% even when noise levels reach 25%. This robustness is attributed to several key mechanisms: the multi-head attention mechanism provides redundant feature protection, the incorporation of graph structure features offers a comprehensive global context.

Figure 6 illustrates the trade-off between prediction lead time (in minutes) and prediction accuracy (%) for the HGTT model compared to traditional methods, such as GCN and LSTM. The chart presents the accuracy of each method as a function of lead time, with the accuracy values shown for different lead times. Traditional methods show limited performance improvements under shorter lead times. For example, SVM and RF exhibit accuracies around 88% to 89% even with shorter lead times, and their accuracy

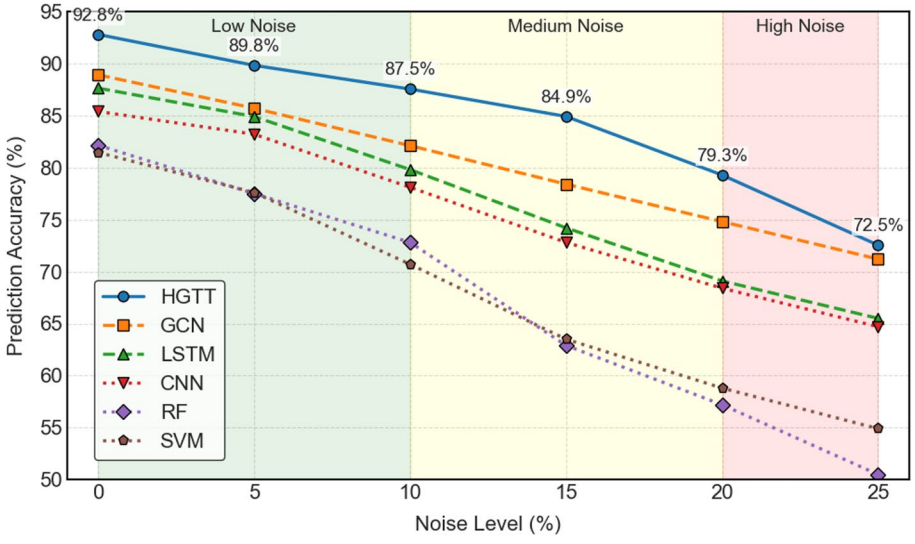


Fig. 5 Model performance under different noise levels

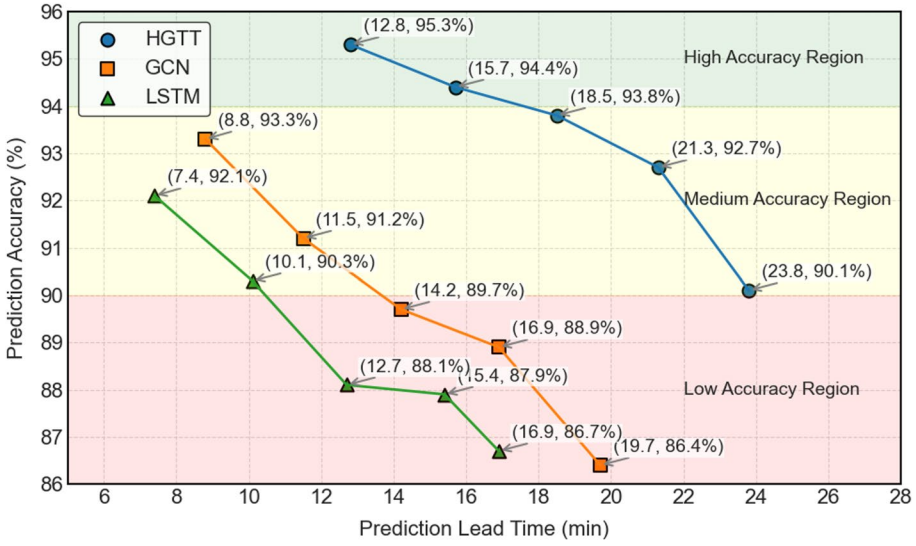


Fig. 6 Trade-off between lead time and accuracy

does not increase significantly when the lead time extends. This is primarily due to their limited ability to utilize early fault indicators and insufficient modeling of dynamic system characteristics. In contrast, the HGTT model demonstrates a superior trade-off between lead time and accuracy. At a lead time of 12.8 min, HGTT achieves an accuracy of 95.3%, and at 21.3 min, it maintains a high accuracy of 92.7%. These results are a direct consequence of the model's causal attention mechanism, which effectively captures long-term dependencies and enhances the model's sensitivity to early fault signs. The integration of graph structure features allows the HGTT model to predict fault propagation trends, while the electrical distance mechanism improves prediction reliability by modeling the physical spread of faults accurately.

6 Conclusion

This paper presents HGTT, a novel deep learning model designed to predict cascading faults in AC/DC hybrid power grids. By effectively integrating GAT and Temporal Transformers, the proposed model captures both spatial and temporal dependencies within the grid, improving prediction accuracy and reducing the need for extensive labeled data. Experimental results demonstrate that HGTT outperforms existing methods, achieving over 15% improvement in fault prediction accuracy while reducing the amount of labeled data required by 50%. More specifically, HGTT achieves a maximum absolute improvement of 14.8% compared with SVM, while reducing labeled data requirements by 50% under the self-supervised pre-training setting. However, challenges remain in large-scale grid systems with high node and edge densities, which may impact the scalability of the model. Additionally, the model assumes a relatively idealized grid structure and may face difficulties when applied to interconnected real-world grids with noise and incomplete data. Furthermore, real-time fault prediction requirements pose additional challenges, as the computational efficiency and scalability of the model must be optimized for faster response in practical applications. Future work could focus on extending the HGTT framework to other power system types, such as microgrids or hybrid renewable systems. It could also improve the scalability of the model by exploring advanced sampling techniques and optimizing the computational efficiency of the graph

and temporal modules. In addition, the node feature cache mechanism is calibrated under quasi-static operating conditions. During rapid topology changes, the cached features may not reflect the updated grid state before the next cache refresh cycle. In such scenarios, the model may produce stale or inaccurate probability estimates until the cache is updated. Future work should investigate adaptive cache invalidation strategies or event-triggered update mechanisms to mitigate this issue in highly dynamic grids.

Nomenclature

A	Actual adjacency matrix element
$\hat{A}$	Predicted adjacency matrix element
α_{ij}	Attention coefficient between nodes v_i and v_j
β	Hyperparameter for electrical distance weight function
b_1, b_2	Bias vectors
c_v^t	Cached feature of node v at time t
d	Dimension of fused edge features
D	Historical operation dataset
ε	Set of edges in the power grid graph
e_{ij}	Feature vector of edge between nodes i and j
G	Power grid graph
$g(\cdot)$	Attention fusion function
$h_i^{(l)}$	Hidden state of node v_i at layer l
h_s	Node feature at current time step
I	Identity matrix
K	Number of attention heads
L	Combined loss function
l	Layer index
λ	Weight coefficient
M_{ij}	Causal mask element
μ	Mean of feature values
N	Total number of nodes
N_i	Neighborhood set of node v_i
P	Active power
p_i	Fault probability for node v_i
θ	Phase angle
S_v^t	Temporal dynamic features of node v at time t
σ	Standard deviation of feature values
T	Number of time steps
U	Voltage amplitude
V	Set of nodes in the power grid graph
X_t	Operational state at time t
y_t	Fault labels at time t
Z	Fused feature
Z_{ij}	Electrical distance between nodes v_i and v_j

Abbreviations

AC	Alternating current
CNN	Convolutional neural network
DC	Direct current
GAT	Graph attention network
GCN	Graph convolutional network
HGTT	Hybrid graph-temporal transformer
HVDC	High voltage direct current
IEEE	Institute of electrical and electronics engineers
IoT	Internet of things
KNN	K-nearest neighbors
LCC	Line commutated converter
LSTM	Long short-term memory
MLP	Multi-layer perceptron
MME	Multi-model ensemble
RF	Random forest
RNN	Recurrent neural network
SVM	Support vector machine

Acknowledgements
Not Applicable.

Author contributions
Wenchao Qin: Methodology, Resources, Writing - Review & Editing.Xufei Liu: Investigation, Conceptualization, Formal Analysis, Writing - Original Draft.Yiran Tao: Investigation, Conceptualization, Supervision, Writing - Original Draft.

Funding
No Funding.

Data availability
The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 28 January 2026 / Accepted: 17 April 2026
Published online: 30 April 2026

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